Exploring Semantic Hadith Overlap Across Topics

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Abstract. Semantic sentence similarity measures the degree of resemblance between multiple sentences. This similarity is a foundational element in information retrieval, machine translation, etc. This paper focuses on natural language processing techniques to analyze the semantic similarity in Hadiths, which are significant religious texts in Islam. Our objective is to investigate the extent of semantiv overlap between Hadiths across various topics, with the aim to provide insights into the cohesion and interconnectedness of Hadiths. We use AraVec and GPT embeddings to represent Hadiths numerically, followed by UMAP (Uniform Manifold Approximation & Projection) to project these embeddings to 2D. The projection serves to visually interpret the relationships between Hadiths, facilitating a deeper understanding of content and semantic interrelations. Our results unveil semantic clusters and connections within Hadiths, contributing to the exploration of Islamic textual heritage through modern computational methodologies. This study suggests that GPT outperforms AraVec, providing a more advanced representation that discerns intricate semantic relationships and subtle nuances within the Hadiths.

Keywords: Hadith · Hadith Corpus · Semantic relatedness · Arabic Natural Language Processing · AraVec & GPT · UMAP

1 Introduction

According to Islamic tradition, a Hadith refers to the recorded sayings and actions of Prophet Muhammad, peace be upon him (pbuh). This literature is considered as the second major source of religious law and moral guidance after the Qur'an [1]. Hadith are composed of two key components: Sanad (سند), the chain

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of narrators who report, and a matn (متن), the actions of Prophet Mohammed (pbuh) [2]. Hadith studies have been a popular subject for Islamic scholars and researchers, who seek to analyze their contents and uncover profound wisdom! [3]. We aim at discerning shared principles and values, unearthing underlying connections and dependencies. These discoveries could greatly aid scholars and individuals seeking a comprehensive understanding of Hadiths. Our approach measures similarity even across takhreej (clusters of Hadith with the same matn but different sanads) and topics to enable intelligent queries and searches in the setting of a knowledge base system. As data, e.g., commentaries and explanations, expand rapidly, it is increasingly vital to provide intelligent tools to extract knowledge from Islamic resources to discern between authentic and counterfeit information. Traditional methods such as manual analysis and interpretation is often time-consuming, labor-intensive, and susceptible to errors [4]. Thus, natural language processing (NLP) stands out as an excellent candidate to process texts at scale, in particular for Hadith research. NLP has the potential to automate the workflow of scholars. Tasks such as disambiguating the names of narrators, visualizing the propagation of a Hadith through narration trees, and even offering initial assessments of a Hadith's authenticity can benefit from NLP techniques [3]. By leveraging computational linguistics and machine learning, NLP techniques can aid Hadith scientists in more efficient and effective processing, analysis, and understanding of Hadith texts. This union of NLP and Hadith studies unveils numerous possibilities, revolutionizing how researchers interact with these sacred texts and expanding the knowledge obtainable from them.

Contributions. We present a study of semantic overlaps between Hadiths that goes beyond clustering into takhreej. Our method is fully automated and unsupervised, offering ease of use and a reduced need for manual labeling of data. We also demonstrate that similarities based on GPT embeddings, for which Arabic is a relatively low-resource language, can outperform LLMs that have been designed specifically for Arabic such as AraVec or AraBERT. We attribute that to GPT's large number of model parameters and extensive training time. We therefore believe this work to be an important step for information retrieval systems with a potential to suggest further readings to users.

Organization. The rest of the paper is organized as follows. In the next section, we provide a background of linguistics (cf. Sect. 2). We then review related work, with a focus on Arabic NLP. We then describe the Hadith data that has been used, the preprocessing applied to it and the methodology (cf. Sect. 4) followed by a discussion of our results (cf. Sect. 5) before concluding and providing potential directions for future research (cf. Sect. 6).

2 Linguistic Background

Linguistic Characteristics of Hadith Texts. Hadith literature, composed in classical Arabic, exhibits unique linguistic features like a specialized lexicon, formulaic expressions (e.g., قال رسول الله - "The Prophet said"), and syntactic struc-

tures distinct from modern Arabic. Grasping these traits is crucial for effective semantic analysis [5].

Embedding Models and Linguistic Nuances. The AraVec model captures semantic relationships based on word co-occurrence within a window, but its static representation might not fully grasp contextual nuances of Hadith texts. In contrast, GPT's transformer architecture excels in contextual understanding, capturing subtleties like different usages of صلاة (prayer) in varied contexts [6].

Semantic Similarity and Linguistic Contexts. Analyzing semantic similarity through embeddings involves understanding polysemy (e.g., نية (intention) having multiple meanings) and synonymy, which is critical for determining semantic similarity. GPT's dynamic embeddings offer a nuanced understanding of such terms in varied contexts compared to AraVec's static embeddings [7].

Semantic Overlap Across Topics. Despite addressing diverse topics, Hadiths exhibit semantic overlap reflecting unified Islamic principles and values. For example, لا تحقرن من المعروف شيئا (Do not belittle any good deed) and إن الله كتب الحسنات والسيئات (Verily, Allah has recorded the good and the bad deeds) converge on the significance of actions, demonstrating semantic linkage captured by embeddings [8].

Common Lexical and Conceptual Foundations. Certain recurring words (إِيمَان - faith), phrases, and concepts permeate through various Hadith topics, creating an interconnected web of meanings and associations. For example, لَيْسَ...الْمُؤْمِنُ بِالطَّعَّانِ and لَا يَزْنِي الزَّانِي... share the concept of "faith" despite addressing different themes [5].

Contextual and Pragmatic Considerations. While lexical and conceptual overlaps contribute to interconnectedness, contextual and pragmatic factors also determine meaning nuances. GPT's contextual embeddings account for linguistic context, capturing subtle differences like من غش فليس منا referring to deception towards the Muslim community versus من غشنا فليس منا having a more general scope [7].

Linguistic Interpretation of UMAP Projections. The UMAP projections visually represent semantic relationships, revealing how different topics or themes cluster together based on linguistic and thematic overlap, offering insights into the interconnections within Hadith literature [6].

3 Arabic Natural Language Processing

NLP has been applied to different languages, including Arabic [9]. ANLP (Arabic Natural Language Processing) is a major research challenge. This is both due to the complex and rich grammar of Arabic, as well as the fact that Arabic is the most-spoken semitic language. Arabic is a complex language, with many

dialects. Working with Arabic text in NLP projects, there are at least four unique challenges [9].

1. Orthographic Ambiguity: The form of characters and spelling of words can vary depending on their context.
2. Morphological Richness: The same verb can have thousands of different forms.
3. Dialectal Variation: There are many dialects of Arabic and there are big differences between them.
4. Orthographic Inconsistency: Since Arabic is a phonetic language, there can be different ways to write the same word when writing in dialectal Arabic, for which there is no agreed-upon standard.

These four different characteristics of Arabic all contribute to data sparsity, with estimates that a full NLP vocbulary would contain millions of words [9]. For example, a single verb in Arabic can have up to 5,400 forms [10]. Before dealing with Arabic words or sentences, texts are usually pre-processed using stemming or lemmatization to mitigate such challenges [11]. To this end, full Arabic text processing tool kits have been developed over the decade, e.g., FARASA by Qatar Computing Research Institute (QCRI) [12], Stanford's Arabic Parser and Word Segmenter [13], etc. In this work, we are concerned only with classical Arabic and a very specific vocabulary and grammar subset. We therefore chose to specialize pre-existing and more general language processing tools to this task to achieve better accuracy, robustness, and reliability.

4 Materials and Methods

4.1 Hadith Dataset

We have chosen the LK-Hadith-Corpus [14] because it provides enough data for our NLP tasks until our work in progress, the completion of a research-specific and clean data set, is completed. In this paper, we consider only the Arabic Hadiths from Sahih al-Bukhari [15], a collection of Hadith compiled by Imam Muhammad al-Bukhari (died 256 AH/870 AD) (pbuh). This collection is considered the most authentic by the majority of the Muslim world and contains over 7,500 hadiths (with repetitions), organized into 97 books, mostly by topic.

4.2 Hadith Data Preprocessing

We first extracted the Arabic matns from the LK-Hadith Corpus [14], specifically from the Sahih Al-Bukhari collection [15]. A new subset was created, focusing on two significant topics: كتاب الصلاة (Prayers, "Salat") and كتاب الوضوء (Ablutions "Wudu"). The goal was to analyze potential overlaps within these topics, with plans to expand the scope to other relevant topics over time. The subset consisted of 282 Hadiths, 168 from كتاب الصلاة and 114 from كتاب الوضوء . We then follow pre-processing with data cleaning to reduce noise and gain reliable insights. General Arabic stopwords defined in the NLTK Arabic library,

Non-Arabic characters, digits, whitespaces, and tashkeel (diacritics indicating Arabic pronunciation) were removed. Post-cleaning, the araby.tokenize() tokenizer from the Python PyArabic package split the corpus into words, with only words longer than three characters retained. Figure 1 shows preprocessing steps that were applied to the LK Hadith dataset.

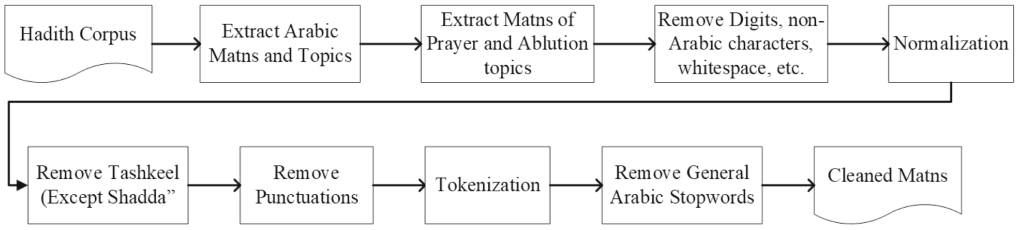

Fig. 1. Hadith Data Preprocessing.

4.3 Methods

Our approach combines text embeddings generated using AraVec [16] and Generative Pre-trained Transformers (GPT) [17] and dimensionality reduction using Uniform Manifold Approximation and Projection (UMAP) [18]. The rationale is to provide a unsupervised analytics pipeline that allows for visual inspection of the semantic overlap of Hadiths across different topics. At the beginning of our pipeline stands a well-structured corpus of Hadith literature which encompasses the Hadith texts sourced from the six canonical collections. Although there are several studies in the Hadith literature, the vast and diverse nature of the texts combined with their inconsistent structure presents challenges in preparing such a well-structured corpus [14].

We use a two-step methodology for this task. Initially, we transformed the matn in the Hadiths into to a vector space using either AraVec [16] or OpenAI's GPT embedding [17] methods. Subsequently, we projected the embedded Hadiths onto a two-dimensional UMAP visualization [18]. We also carried out a comparative analysis of the results from both embedding methods, aiming to identify the more effective approach. We will then use this analysis to guide our choice of the best embedding model for future work for developing a knowledge base system. AraVec, an Arabic adaptation of word2vec, is an open-source initiative offering pre-trained distributed word representations, or word embeddings. Its primary purpose is to equip the Arabic NLP research community with freely accessible and potent word embedding models. These models prove invaluable for several NLP tasks involving Arabic text analysis, allowing researchers and practitioners to harness the advantages of word embeddings in their Arabic language applications. The generative pre-trained transformer (GPT) is a cutting-edge language model created and maintained by OpenAI. As part of the neural network models' family, it utilizes a transformer architecture based on a self-attention mechanism [19]. This feature enables the model to capture relationships between words or tokens within a text. GPT represents a significant breakthrough in artificial intelligence, underpinning generative AI applications such as

ChatGPT. GPT was trained on vast online text data, learned word semantics, grammar, and patterns, enriching its language understanding and generation abilities. OpenAI also provides text embedding models, converting text strings into embedding vectors, or vectorized data preserving content meaning—similar input is expected to produce similar embeddings. These models, which input text strings and output embedding vectors, assess text relatedness effectively. Closer distances indicate higher relatedness, and vice versa. These embeddings benefit search, clustering, recommendations, anomaly detection, diversity measurement, and classification. To generate an embedding, input the text to the embeddings API endpoint with the chosen model ID, such as "text-embedding-ada-002". The resulting API [20] provides the necessary embedding for subsequent processes. Overall, GPT's language modeling advances and OpenAI's text embedding models have significantly transformed AI applications, offering superior text generation, understanding, and analysis capabilities. UMAP [18] is a powerful, scalable dimensionality reduction algorithm and data analysis tool. It competes well with advanced methods like t-distributed stochastic neighbor embedding (t-SNE) [21] and is popular for unsupervised clustering. UMAP addresses the "curse of dimensionality" by identifying clusters in high-dimensional space and retaining both intra- and inter-cluster distances (unlike t-SNE which can struggle with inter-cluster distances) in lower-dimensional embeddings. In addition, it matches the speed of Principal Component Analysis (PCA) [22], making it significantly faster than t-SNE. Figure 2 shows the workflow that has been followed in this study for semantic overlap of Hadith across topics.

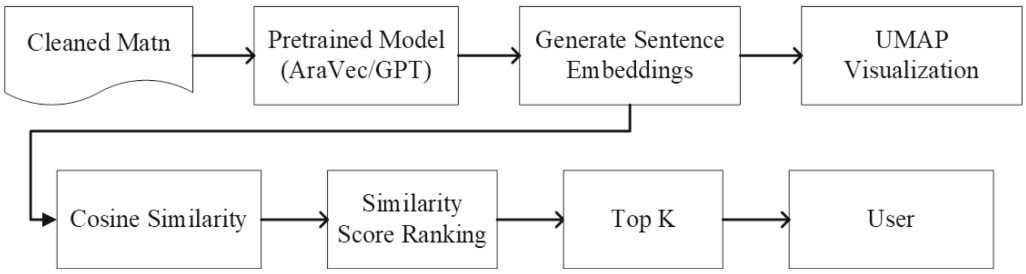

Fig. 2. Workflow adopted in this study.

5 Results and Discussion

The cleaned corpus totaled 60,311 words from the subset covering the chosen topics from the Sahih Al-Bukhari collection, with a vocabulary of 35 unique words. The longest single matn in the subset had 236 words. Figure 3 shows a word cloud of matns for the topics كتاب الصلاة and كتاب الوضوء . As can be seen, several terms are closely linked to ablution and prayer, such as الصلاة (prayer), المسجد (mosque), الكعبة (Kaaba), البيت (house), الجنة (paradise), القبلة (qibla), فتوضأ (perform ablution), فغسل وجهه (washing the face), ركعتين ركعتين (two units of prayer), الوضوء (ablution),

Fig. 3. Word cloud for topics كتاب الصلاة and كتاب الوضوء .

وجهه ثلاثا (wiping the face three times), فمسح (wiping), والعصر (Asr prayer), and المغرب (maghrib prayer). These terms are essential within the context of ablution and prayer, representing crucial elements, actions, and locations related to these practices. In Fig. 3, some words are repeated with different weights. This is due to the collocations, combination of words that frequently occur together and convey specific meanings or cultural connotations. These combinations often consist of multi-word expressions or phrase that are deeply rooted in Arabic culture, religion, literature and everyday communication. For example صلى الله عليه و سلم ("peace be up on him") is a phrase or a multi word expressions and the word cloud generator will attempt to identify and include these collocations in the word cloud. In the future, we plan to detect these collocations and remedy this effect. After preprocessing the matn in the data set, the subsequent step involved embedding the matn using the AraVec and GPT models, ensuring that similar documents clustered closely. These embeddings could be utilized for other tasks such as corpus visualization or running clustering algorithms like KMeans or DBSCAN [23]. To check sentence overlap and compare the mentioned embedding models, we initially embedded the matn using the AraVec model. We loaded the AraVec model with the Python "gensim" module to generate high-dimensional vector word embeddings, placing semantically similar words close to each other in the vector space. We computed sentence embeddings by averaging word embeddings in each sentence. We embedded the matn using the GPT model by submitting each matn to the embeddings API endpoint with the embedding model ID. Specifically, we used the second-generation ada model (text-embedding-ada-002) from OpenAI GPT's four base embedding models (davinci, curie, ada, and babbage). This model, simpler, cheaper, and faster, produces 1,024 dimensions and outperforms older models on text search, code search, and sentence similarity tasks. After obtaining the data set's matn embeddings, we saved the data set and embeddings in a CSV file for future use. The next step involved

Table 1. High similarity between matns using GPT and AraVec.

Nº	Matn	Topic	GPT	AraVec
1	خَرَجَ رَسُولُ اللَّهِ صلى الله عليه وسلم بِالْهَاجِرَةِ فَصَلَّى بِالْبَطْحَاءِ الظُّهْرَ وَالْعَصْرَ رَكْعَتَيْنِ، وَنَصَبَ بَيْنَ يَدَيْهِ عَنَزَةً، وَتَوَضَّأَ، فَجَعَلَ النَّاسُ يَتَمَسَّحُونَ بِوَضُوئِهِ.	كتاب الصلاة	0.95	0.91
2	خَرَجَ عَلَيْنَا رَسُولُ اللَّهِ صلى الله عليه وسلم بِالْهَاجِرَةِ، فَأُتِيَ بِوَضُوءٍ فَتَوَضَّأَ، فَجَعَلَ النَّاسُ يَأْخُذُونَ مِنْ فَضْلِ وَضُوئِهِ فَيَتَمَسَّحُونَ بِهِ، فَصَلَّى النَّبِيُّ صلى الله عليه وسلم الظُّهْرَ رَكْعَتَيْنِ وَالْعَصْرَ رَكْعَتَيْنِ، وَبَيْنَ يَدَيْهِ عَنَزَةٌ.	كتاب الوضوء	0.91	0.91

Table 2. Low similarity between matns using GPT and AraVec.

Nº	Matn	Topic	GPT	AraVec
1	أَنَّ زَيْدَ بْنَ خَالِدٍ، أَرْسَلَهُ إِلَى أَبِي جُهَيْمٍ يَسْأَلُهُ مَاذَا سَمِعَ مِنْ، رَسُولِ اللَّهِ صلى الله عليه وسلم فِي الْمَارِّ بَيْنَ يَدَيِ الْمُصَلِّي فَقَالَ أَبُو جُهَيْمٍ قَالَ رَسُولُ اللَّهِ صلى الله عليه وسلم لَوْ يَعْلَمُ الْمَارُّ بَيْنَ يَدَيِ الْمُصَلِّي مَاذَا عَلَيْهِ لَكَانَ أَنْ يَقِفَ أَرْبَعِينَ خَيْرًا لَهُ مِنْ أَنْ يَمُرَّ بَيْنَ يَدَيْهِ . قَالَ أَبُو النَّضْرِ لاَ أَدْرِي أَقَالَ أَرْبَعِينَ يَوْمًا أَوْ شَهْرًا أَوْ سَنَةً.	كتاب الصلاة	0.76	0.76
2	سَمِعْتُ النَّبِيَّ صلى الله عليه وسلم يَقُولُ `` لاَ يَتَوَضَّأُ رَجُلٌ فَيُحْسِنُ وُضُوءَهُ، وَيُصَلِّي الصَّلاَةَ إِلاَّ غُفِرَ لَهُ مَا بَيْنَهُ وَبَيْنَ الصَّلاَةِ حَتَّى يُصَلِّيَهَا ". قَالَ عُرْوَةُ الآيَةُ إِنَّ الَّذِينَ يَكْتُمُونَ مَا أَنْزَلْنَا مِنَ الْبَيِّنَاتِ.	كتاب الوضوء	0.75	0.79

fitting the UMAP embedding to the matn embeddings from both models and examining the resulting two-dimensional embedding. We used default values for UMAP's parameters (min_dist=0.1, n_neighbors=15, and metric="cosine"). UMAP's min_dist parameter defines the minimum distance between points in the low-dimensional embedding space, controlling the visualization's granularity. Generally, these values balance global structure preservation and local relationship capture. Adjusting the min_dist value can fine-tune visualization based on data characteristics. In UMAP, we selected the cosine metric, which measures vector similarity or dissimilarity by computing the angle between them, especially beneficial for high-dimensional or vector-represented text data. For two points $\mathbf{x}, \mathbf{y}$ embedded onto a unit hyper-sphere, maximizing the cosine norm is equivalent to minimizing the Euclidean distance, but is faster to compute:

$$(\mathbf{x}-\mathbf{y})^2 = \underbrace{\mathbf{x}^T\mathbf{x}}_{=1} + \underbrace{\mathbf{y}^T\mathbf{y}}_{=1} - 2\langle \mathbf{x}, \mathbf{y} \rangle = 2\left(1 - \underbrace{\langle \mathbf{x}, \mathbf{y} \rangle}_{\cos \angle \mathbf{x},\mathbf{y}}\right), \tag{1}$$

Common alternatives include Manhattan, and correlation distances. The chosen metric depends on data nature and analysis requirements. In our case, to identify the most similar matns from given topics, the cosine metric best computed the distance. We perform ranking followed by a selection of the top-k (e.g., $k = 10$) most similar matns. This allows users to select further readings from suggested Hadith. Tables 1 and 2 show top two high and low similar matns (respectively) using cosine similarity for the given matn ``خَرَجَ عَلَيْنَا رَسُولُ اللَّهِ صلى الله عليه وسلم بِالْهَاجِرَةِ، فَأُتِيَ بِوَضُوءٍ فَتَوَضَّأَ فَصَلَّى بِنَا الظُّهْرَ وَالْعَصْرَ وَبَيْنَ يَدَيْهِ عَنَزَةٌ، وَالْمَرْأَةُ وَالْحِمَارُ يَمُرُّونَ مِنْ وَرَائِهَا."

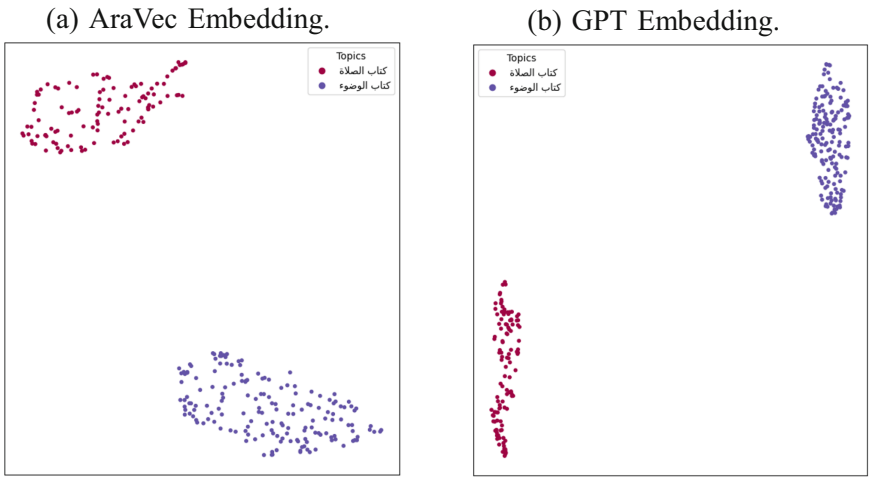

Fig. 4. Supervised UMAP reduction to 2D AraVec vs. GPT embeddings.

UMAP's n_neighbors controls the number of neighboring points used to construct the neighborhood graph of the manifold in the high-dimensional space (perplexity) and affects local structure preservation and the low-dimensional embedding's overall fidelity. Higher values provide a more global data view, capturing broader relationships between high-dimensional space points. Conversely, lower values focus on more local relationships and finer data details. The appropriate value depends on data set characteristics and the desired embedding granularity. Generally, smaller data sets may require a lower n_neighbors value (5–50), while larger data sets may benefit from a higher values (50–200). Different values are often tested to identify the setting that best represents the data's underlying structure in the low-dimensional embedding. Figure 4 shows the resulting UMAP embeddings for AraVec and GPT embeddings.

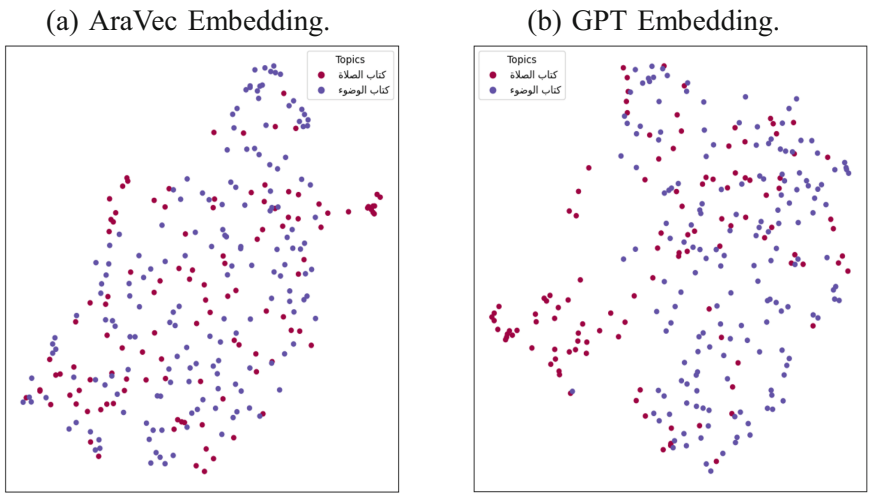

Fig. 5. Unsupervised UMAP reduction to 2D of AraVec vs. GPT embeddings.

Figure 5 shows the UMAP visualization of *unsupervised* clustering, that means clustering based only on the similarity of the matns in the data set. Figure 4 presents a *supervised* UMAP, exploiting both the matns' embeddings as well as their class labels. Since supervised UMAP uses class labels directly in the dimension reduction process, the result is a low-dimensional projection that not only retains the intrinsic data structure but also the class information [24]. In contrast, Fig. 5 depicts an unsupervised UMAP projection without class labels. The fitting and transformation occur solely due to the data's intrinsic structure, enabling the algorithm to reveal patterns and relationships without prior class data [24]. From this result, it is evident that there is significant overlap in the matns, as suggested by the similarity scores. This figure allows identification of Hadith clusters with substantial semantic overlap, signifying a specific interrelation between prayer and ablution topics. Furthermore, it is apparent from these figures that GPT embeddings surpass AraVec embeddings. GPT embeddings deliver a superior data representation, yielding more insightful clusters and enhanced visualization. This can be seen in Fig. 4 by the more localized clusters, and implies that the GPT embedding model more effectively discerns semantic relationships and subtleties within the Hadiths than the AraVec model.

6 Conclusion and Future Work

The comparison drawn from these visual representations indicates the superiority of GPT embeddings over AraVec embeddings. The evidence suggests that GPT embeddings capture a more better and refined representation of the data, resulting in the creation of clusters that offer deeper insights and a more enriched visual interpretation. In particular, GPT clusters in Fig. 4 show better localization and even the unsupervised case (cf. Fig. 5) shows better topic separation to the left of the figure. These outcome suggests that the GPT embedding model outperforms the AraVec model in its ability to accurately discern intricate semantic relationships and subtle nuances present within the Hadiths. In the future, we will continue our ongoing efforts together with Islamic scholars to proof-read and cleanse the data, and to obtain an authoritative ground truth of similarities between Hadith. This goes beyond the clustering of takhreej, which offers variations in the chains of narrators of the same Hadith. At the same time, we are exploring how textual similarity measures, such as presented in this work, can be used for presenting recommended Hadiths to readers, primarily laypersons, who seek topic-based guidance from the Hadith. We will make the resulting data set containing the embeddings available to the public for further academic dissemination in the future.

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References

1. Sulisto, B., Ramadhan, A., Abdurachmal, E., Zarlis, M., Trisetyarso, A.: The utilization of machine learning on studying hadith in Islam: a systematic literature review. Educ. Inf. Technol. (2023). https://doi.org/10.1007/s10639-023-12008-9
2. Mughal, M.A.: Ibtidā' Al-sanad/إِبْتِدَاء اَلْسَنَد [Beginning of the chain of narrators of Ḥadīth] [a term of science of Ḥadīth]. SSRN (2016). https://ssrn.com/abstract=2717633
3. Azmi, A., Al-Qabbany, A., Hussain, A.: Computational and natural language processing based studies of hadith literature: a survey. Artif. Intell. Rev. **52**, 1369–1414 (2019). https://doi.org/10.1007/s10462-019-09692-w
4. Najeeb, M.: Towards innovative system for hadith isnad processing. Int. J. Comput. Trends Technol. **18**, 257–259 (2014). https://doi.org/10.14445/22312803/IJCTT-V18P154
5. Al-Azawei, R.: Linguistic features of hadith texts for authenticity: a machine learning approach. Int. J. Islamic Arabic Stud. **9**(3), 1–10 (2019). ISSN 2520-2218. https://doi.org/10.1186/s40092-019-00204-6
6. Al-Azawei, R.: A comprehensive study on the use of hadith in learning Arabic as a second language. Int. J. Islamic Arabic Stud. **10**(1), 1–13 (2020). ISSN 2520-2218. https://doi.org/10.1186/s40092-020-00219-3
7. Al-Azawei, R.: A survey of computational and natural language processing-based studies of hadith literature. Int. J. Islamic Arabic Stud. **9**(2), 1–12 (2019). ISSN 2520-2218. https://doi.org/10.1186/s40092-019-00203-7
8. Al-Azawei, R.: Translation of hadiths into english: challenges and opportunities. Int. J. Islamic Arabic Stud. **9**(4), 1–8 (2019). ISSN 2520-2218. https://doi.org/10.1186/s40092-019-00205-5
9. Bourahouat, G., Abourezq, M., Daoudi, N.: Systematic review of the Arabic natural language processing: challenges, techniques and new trends. J. Theor. Appl. Inf. Technol. **101**(3), 1333–1343 (2023). http://www.jatit.org/volumes/hundredone3.php
10. Alsayed, A., Qadah, T.M., Arif, M.: A performance analysis of transformer-based deep learning models for Arabic image captioning. J. King Saud Univ.-Comput. Inf. Sci. **35**(9), 101750 (2023)
11. Oueslati, O., Cambria, E., HajHmida, M.B., Ounelli, H.: A review of sentiment analysis research in Arabic language. Future Gener. Comput. Syst. **112**, 408–430 (2020)
12. Abdelali, A., Darwish, K., Durrani, N., Mubarak, H.: Farasa: a fast and furious segmenter for Arabic. In: Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations, pp. 11–16 (2016). https://doi.org/10.18653/v1/N16-3003
13. Green, S., Manning, C.D.: Better Arabic parsing: baselines, evaluations, and analysis. In: Proceedings of the 23rd International Conference on Computational Linguistics, pp. 394–402 (2010). https://aclanthology.org/C10-1045
14. Altammami, S., Atwell, E., Alsalka, A.: The Arabic-English parallel corpus of authentic hadith. Int. J. Islamic Appl. Comput. Sci. Technol. (IJASAT) 1–10 (2019). http://www.sign-ific-ance.co.uk/index.php/IJASAT/article/view/2199/1908

15. Muḥammad ibn Ismāʿīl Al-Bukhārī. الجامع المسند الصحيح المختصر من أُمور رسول الله الجامع المسند الصحيح المختصر من أُمور رسول الله (The shortened authentic collection with isnads from the affairs of the messenger, peace be upon him, and his traditions and his days). ISBN: 978-1-56744-519-0. 846
16. Soliman, A.B., Eissa, K., El-Beltagy, S.R.: Aravec: a set of Arabic word embedding models for use in Arabic NLP. Procedia Comput. Sci. **117**, 256–265 (2017). https://doi.org/10.1016/j.procs.2017.10.117
17. OpenAI. GPT-4 technical report (2023). arXiv:2303.08774v3
18. McInnes, L., Healy, J., Melville, J.: UMAP: uniform manifold approximation and projection for dimension reduction. arXiv:1802.03426v3 (2023)
19. Vaswani, A., et al.: Attention is all you need. In: Advances in Neural Information Processing Systems, vol. 30 (2017)
20. Keita, Z.: Introduction to text and code embeddings in the openai API (2023). https://www.datacamp.com/tutorial/introduction-to-text-embeddings-with-the-open-ai-api. Accessed 8 Oct 2023
21. van der Maaten, L.J.P., Hinton, G.E.: Visualizing data using t-SNE. J. Mach. Learn. Res. **9**(86), 2579–2605 (2008). http://jmlr.org/papers/v9/vandermaaten08a.html
22. Pearson, K.: On lines and planes of closest fit to systems of points in space. Phil. Mag. **2**(11), 559–572 (1901). https://doi.org/10.1080/14786440109462720
23. Reades, J., Williams, J.: Clustering and visualising documents using word embeddings (2023). https://programminghistorian.org/en/lessons/clustering-visualizing-word-embeddings. Accessed 8 Oct 2023
24. Ghosh, T., Kirby, M.: Supervised dimensionality reduction and visualization using centroid-encoder. J. Mach. Learn. Res. **23**(1), 901–934 (2022). https://jmlr.csail.mit.edu/papers/v23/20-188.html